10/06/2026

Code-A Phase-1

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027

FT - 01 (RM)

MM: 720 **Time: 180 Mins.**

Answers

1. (1)	37. (4)	73. (2)	109. (3)	145. (2)
2. (2)	38. (3)	74. (1)	110. (4)	146. (2)
3. (2)	39. (2)	75. (3)	111. (3)	147. (1)
4. (1)	40. (4)	76. (2)	112. (3)	148. (4)
5. (3)	41. (2)	77. (4)	113. (3)	149. (4)
6. (2)	42. (1)	78. (3)	114. (2)	150. (4)
7. (1)	43. (4)	79. (2)	115. (4)	151. (3)
8. (4)	44. (2)	80. (2)	116. (3)	152. (4)
9. (2)	45. (1)	81. (4)	117. (4)	153. (2)
10. (2)	46. (4)	82. (4)	118. (4)	154. (2)
11. (3)	47. (1)	83. (4)	119. (3)	155. (1)
12. (4)	48. (2)	84. (1)	120. (4)	156. (1)
13. (2)	49. (4)	85. (1)	121. (2)	157. (3)
14. (3)	50. (2)	86. (3)	122. (3)	158. (4)
15. (3)	51. (3)	87. (3)	123. (2)	159. (1)
16. (2)	52. (3)	88. (1)	124. (2)	160. (1)
17. (4)	53. (3)	89. (3)	125. (3)	161. (1)
18. (2)	54. (2)	90. (4)	126. (3)	162. (4)
19. (3)	55. (3)	91. (2)	127. (2)	163. (4)
20. (2)	56. (1)	92. (3)	128. (3)	164. (2)
21. (1)	57. (1)	93. (4)	129. (1)	165. (4)
22. (2)	58. (4)	94. (1)	130. (4)	166. (4)
23. (1)	59. (3)	95. (3)	131. (3)	167. (2)
24. (3)	60. (4)	96. (2)	132. (2)	168. (2)
25. (1)	61. (2)	97. (2)	133. (2)	169. (3)
26. (4)	62. (4)	98. (2)	134. (4)	170. (3)
27. (3)	63. (2)	99. (2)	135. (3)	171. (3)
28. (1)	64. (2)	100. (2)	136. (2)	172. (4)
29. (1)	65. (3)	101. (3)	137. (1)	173. (4)
30. (3)	66. (1)	102. (2)	138. (4)	174. (1)
31. (3)	67. (4)	103. (2)	139. (1)	175. (4)
32. (2)	68. (1)	104. (3)	140. (2)	176. (2)
33. (4)	69. (2)	105. (4)	141. (3)	177. (2)
34. (4)	70. (2)	106. (2)	142. (2)	178. (4)
35. (4)	71. (1)	107. (2)	143. (2)	179. (4)
36. (4)	72. (4)	108. (4)	144. (4)	180. (2)

Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:
$\frac{\Delta R}{R} = 2\frac{\Delta a}{a} + \frac{1}{2}\frac{\Delta b}{b} + 4\frac{\Delta c}{c} + \frac{\Delta d}{d}$

(2) **Answer :** (2)
Solution:
Here dropped digit is 5 and preceding digit is even (4) remain unchanged.

(3) **Answer :** (2)
Solution:
Percentage error in $V = \frac{4}{80} \times 100 = 5\%$
Percentage error in $I = \frac{0.1}{10} \times 100 = 1\%$
Thus, percentage error in R = 5% + 1% = 6%

(4) **Answer :** (1)
Hint:
L.C = 1 MSD – 1 VSD
Solution:
L.C = 1 MSD – 1 VSD
$= MSD\left\{1 - \frac{VSD}{MSD}\right\}$
1 MSD = 1 mm
19 MSD = 20 VSD
1 VSD = $\frac{19}{20}VSD$
L.C = $1 \times \left\{1 - \frac{19}{20}\right\} = 0.05$ mm

(5) **Answer :** (3)
Hint:
Non-zero digits are significant.
Solution:
Trailing zeros in a number without decimal are insignificant.

(6) **Answer :** (2)
Solution:
A must be most accurate, therefore option (2) and (3) are preferable where B must be most precise therefore from options, (2) is preferable.
Therefore answer is (2)

(7) **Answer :** (1)
Hint:
Absolute mean error $= \frac{\text{Sum of error in individual observation}}{\text{Number of observation}}$
Solution:
Mean reading $= \frac{1.25+1.24+1.27+1.21+1.28}{5} = 1.25$
Δ_1 = 1.25 – 1.25 = 0, Δ_2 = 0.01, Δ_3 = 0.02
Δ_4 = 0.04, Δ_5 = 0.03
$\Rightarrow \Delta_{mean} = \frac{0.01+0.02+0.04+0.03}{5} = 0.02$

(8) **Answer :** (4)
Hint:
$\frac{\Delta T}{T} \times 100\% = \frac{1}{2}\frac{\Delta l}{l} \times 100\% + \frac{1}{2} \times \frac{\Delta g}{g} \times 100\%$
Solution:
$T = 2\pi\sqrt{\frac{l}{g}}$
Squaring

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$T^2 = 4\pi^2 \frac{l}{g} \Rightarrow g = \frac{4\pi^2 l}{T^2}$

$\frac{\Delta g}{g} \times 100\% = \frac{\Delta l}{l} \times 100\% + 2\frac{\Delta T}{T} \times 100\%$

= 3% + 2 × 2% = 7%

(9) **Answer :** (2)
Solution:
For $z = x - y$
⇒ $\Delta z = \Delta x + \Delta y$

(10) **Answer :** (2)
Solution:
Zero error is a systematic error.

(11) **Answer :** (3)
Solution:
Random error $\propto \frac{1}{\text{Number of observation}}$

(12) **Answer :** (4)
Solution:
The magnitude of difference between true value or actual value and measured value in a experiment is called absolute error.

(13) **Answer :** (2)
Solution:
Least count = 1MSD – 1VSD

$= 1 \text{ MSD} - \frac{9}{10}\text{MSD}$

$= \frac{1}{10}\text{MSD}$

$= \frac{1}{10} \times 0.1 \text{ cm} = 0.01 \text{ cm}$

Zero error = +0.1 cm
Main scale reading = 5 cm
Vernier scale reading = 8 × 0.01 = 0.08 cm
Final measurement of diameter
= 5 + 0.08 – 0.1 = 4.98 cm

(14) **Answer :** (3)
Solution:
In multiplication of two physical quantities, there should be as many significant figures retained in final result, as there in the original number with the least significant figures.

(15) **Answer :** (3)
Hint:
sin(ω*t* + *bx*) is dimensionless quantity
Solution:
[*b*] = [L^{-1}]

$\left[\frac{a}{b}\right] = [M^0L^0T^0]$

[*a*] = [*b*] = [$M^0L^{-1}T^0$]

(16) **Answer :** (2)
Solution:
[U] = [ML^2T^{-2}], [B] = [*x*] = [*L*]

$\left[A\right] = \frac{[ML^2T^{-2}][L]}{[L^{1/2}]} = \left[ML^{5/2}T^{-2}\right]$

hence [AB] = [$ML^{7/2}T^{-2}$]

(17) **Answer :** (4)
Solution:
Work and torque have same dimensional formula.
$L = mvr = \frac{nh}{2\pi}$

(18) **Answer :** (2)
Solution:

$\frac{d^3y}{dx^3} = \frac{[ML^{-2}T^{-2}]}{[MLT^{-2}]^3} = [M^{-2}L^{-5}T^4]$

(19) **Answer :** (3)
Solution:
In 2.735 digit to rounded off is odd, so it is increased by 1 and in 2.785 digit to be rounded off is even, so it is left unchanged.

(20) **Answer :** (2)
Solution:
$E \propto M^a A^b V^c$
$[ML^2T^{-2}] = K[M]^a [LT^{-2}]^b [LT^{-1}]^c$
$a = 1$, $b + c = 2$, $-2b - c = -2$
$2b + c = 2$
$b + c = 2$
$b = 0$
$c = 2$

(21) **Answer :** (1)
Solution:
Velocity → $[LT^{-1}]$
Acceleration → $[LT^{-2}]$
Force → $[MLT^{-2}]$
Pressure → $[ML^{-1}T^{-2}]$

(22) **Answer :** (2)
Solution:
The slope of position-time graph which shows velocity in option (2), is decreasing with time. So, acceleration is negative in it.

(23) **Answer :** (1)
Hint:
Change in velocity = Area under *a*-*t* graph
Solution:
$v_{final} - v_{initial} = 20$
$v_f = 23$ m/s

(24) **Answer :** (3)
Solution:
Acceleration (a) = rate of change of velocity w.r.t. time.
$\vec{a} = \frac{d\vec{v}}{dt}$

(25) **Answer :** (1)
Solution:
$a_{avg} = \frac{\Delta v}{\Delta t} = \frac{15-9}{3} = \frac{6}{3} = 2\ \text{m/s}^2$

(26) **Answer :** (4)
Solution:

Time taken by man to go from his home to market $t_1 = \frac{Distance}{Speed} = \frac{3\,km}{6\ km\ h^{-1}} = \frac{1}{2}\,h = 30\ \text{min}.$ Time taken by man to go from market to home $t_2 = \frac{3\,km}{9\ km\ h^{-1}} = \frac{1}{3}\,h = 20\ \text{min}.$

Average speed $= \frac{3+3}{\frac{50}{60}} = 7.2\ \text{km}/h$

Average velocity $= \frac{0}{\frac{50}{60}} = 0$

(27) **Answer :** (3)
Solution:
$t = \frac{t_{train} + l_{platform}}{v}$
$30 = \frac{l+400}{20}$
$l = 200$ m

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(28) **Answer :** (1)
Solution:
$\int_0^{\pi} \sin x dx = [-\cos x]_0^{\pi}$
= – [cos π – cos 0] = – [–1 – 1] = 2

(29) **Answer :** (1)
Solution:
Given line have positive intercept but negative slope. So its equation can be written as
$v = -mx + v_0$ …(i) $\left[\text{When } m = \tan\theta = \frac{v_0}{x_0}\right]$
By differentiating with respect to time we get $\frac{dv}{dt} = -m\frac{dx}{dt} = -mv$
Now substituting the value of *v* from equation (i) we get $\frac{dv}{dt} = -m\left[-mx + v_0\right] = m^2x - mv_0$
$\therefore \quad a = m^2x - mv_0$
i.e. the graph between *a* and *x* should have positive slope but negative intercept on *a*-axis. So graph (1) is correct.

(30) **Answer :** (3)
Solution:
$X = t^3 - 3t^2 + 4$
$\frac{dx}{dt} = 3t^2 - 6t$
$\frac{d^2x}{dt^2} = 6t - 6$
$a = 0$
$6t - 6 = 0$
$t = 1$
$\left[\frac{dx}{dt}\right]_{t=1} = 3 - 6 = -3$
Speed = 3 m/s

(31) **Answer :** (3)
Solution:
$t = \frac{200}{15+5} = 10\,\text{s}.$

(32) **Answer :** (2)
Solution:
Use equation of motion,
$v^2 = u^2 + 2as$
$v^2 = u^2 + 2gh$

(33) **Answer :** (4)
Solution:
At $t = 8$, body landing on the roof
$H = ut + \frac{1}{2}gt^2$
$= 60 \times (8) - \frac{1}{2} \times 10 \times 64$
= 480 – 320 = 160 m

(34) **Answer :** (4)
Solution:
Time = $\frac{\text{Distance}}{\text{Velocity}} = \frac{\text{Total distance travelled}}{\text{Relative velocity}}$
$= \frac{80+120}{(10+15)}\text{m/s}$
$= \frac{200}{25} = 8\,\text{s}$

(35) **Answer :** (4)
Solution:
$t = \frac{t_1 t_2}{t_1 + t_2}$
$= \frac{60\times30}{60+30} = \frac{60\times30}{90} = 20\ s$

(36) **Answer :** (4)
Hint:
$\frac{1}{2}g(4)^2 = \frac{1}{2}g\left[t^2 - (t-2)^2\right]$

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Solution:
Let time of fall be *t* sec. In last two seconds, distance covered
$= \frac{1}{2}gt^2 - \frac{1}{2}g(t-2)^2$
= 5[t^2 – $(t-2)^2$]
= 5[4*t* – 4]
According to question,
5(4*t* – 4) = $\frac{1}{2} \times 10 \times (4)^2$
5(4*t* – 4) = 80
⇒ *t* = $\frac{20}{4}$
= 5 s
Height of tower,
H = $\frac{1}{2} \times 10 \times (5)^2 = 125\,\text{m}$

(37) Answer : (4)
Hint:
For free fall *u* = 0
$h = \frac{1}{2}gt^2$
Solution:
For 1st half,
$t_1 = \sqrt{\frac{h}{g}}$
(t_2 + t_1) is total time of fall
Hence $t_2 + t_1 = \sqrt{\frac{2h}{g}}$
$\frac{t_2+t_1}{t_1} = \sqrt{2}$

(38) Answer : (3)
Solution:
Slope of acceleration vs time graph gives acceleration.
Acceleration $a = \frac{dv}{dt}$

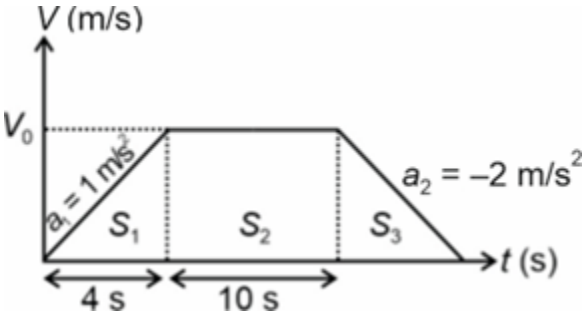
(39) Answer : (2)
Solution:
$v = u + at \Rightarrow 0 = u - \frac{g}{n} \Rightarrow u = \frac{g}{n}$
(time taken by each ball to reach the highest point $t = \frac{1}{n}$)
$v^2 = u^2 - 2gh \Rightarrow h = \frac{u^2}{2g} = \frac{g}{2n^2}$

(40) Answer : (4)
Solution:
Acceleration depends on rate of change of *v* with time.

(41) Answer : (2)
Solution:
$\sqrt{\frac{v^2+u^2}{2}} = \sqrt{\frac{(7u)^2+u^2}{2}} = \sqrt{\frac{50u^2}{2}} = 5u$

(42) Answer : (1)
Hint:
S = S_1 + S_2 + S_3
Solution:

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$S_1 = \frac{1}{2} \times 1 \times 4^2 = 8$ m

$S_2 = v_0 t_2 = (a_1 t_1) t_2 = 1 \times 4 \times 10 = 40$ m

$0 = v_0^2 + 2a_2 S_3$

$S_3 = \frac{16}{2 \times 2} = 4$ m

$S = 40 + 8 + 4 = 52$ m

(43) **Answer :** (4)
Solution:
V = slope of x-t curve

$\frac{V_A}{V_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{3}$

(44) **Answer :** (2)
Hint:
Muzzle velocity of a bullet is the velocity of bullet w.r.t. the gun.
Solution:
Velocity of bullet w.r.t. ground ($v_{b/g}$) is
200 + 80 = 280 m/s
Velocity of bullet as seen by the thief:

$\vec{v}_{b/T} = \vec{v}_{b/g} - \vec{v}_{T/g}$

$v_{b/T} = 280 - 50 = 230$ m/s

(45) **Answer :** (1)
Solution:
Statement-I is true as in case of motion under gravity.
Statement-II is true as in case of accelerating a car then braking.

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CHEMISTRY

(46) **Answer :** (4)
Solution:
1 mol N_2 $\equiv$ 2 mol NO $\equiv$ 2 mol NO_2
$\therefore$ 0.5 mol N_2 $\equiv$ 1 mol NO $\equiv$ 1 mol NO_2
Mass of NO_2 = 1 × 46 = 46 g

(47) **Answer :** (1)
Solution:
mass of Fe = 100 × $\frac{0.33}{100}$ = 0.33 g
$\therefore$ mole of Fe = $\frac{0.33}{56}$ = 5.89×10^{-3} mole
$\therefore$ Number of atoms of Fe = $5.89 \times 10^{-3} \times 6.022 \times 10^{23} = 0.035 \times 10^{23}$ atoms

(48) **Answer :** (2)
Solution:
$n = \frac{1.5055 \times 10^{23}}{6.022 \times 10^{23}} = 0.25$ mol
Mass of 1 mole of CO = 28 g
Mass of 0.25 mol of CO = 7 g

(49) **Answer :** (4)

Hint:
Number of atoms = Moles × N_A × Atomicity
Solution:
Let mass of each of S_2 and S_8 taken be x g

S_2	S_8
Moles = $\frac{x}{32\times2}$	$\frac{x}{32\times8}$
Molecules = $\frac{x}{32\times2}\times N_A$	$\frac{x}{32\times8}\times N_A$

Let mass of each O_2 and O_3 taken be x g
Number of atoms in O_2 molecule $= \frac{x}{16\times2}N_A\times2 = \frac{x}{16}N_A$
Number of atoms in O_3 molecule $= \frac{x}{16\times3}N_A\times3 = \frac{x}{16}N_A$

(50) Answer : (2)
Hint:
1 mole $CaCO_3$ gives 1 mole CO_2 on complete decomposition.
Solution:
$CaCO_3 \rightarrow CaO + CO_2$
$n_{CO_2} = n_{CaCO_3} = \frac{5.6}{22.4} = 0.25$ mole
$w_{CaCO_3} = 0.25\times100 = 25\,g$
% purity = $\frac{25}{150}\times100$ = 16.67%

(51) Answer : (3)
Solution:
Moles of CH_4 = $\frac{1.6}{16}$ = 0.1 mol
Number of electrons of CH_4 = 0.1 × 10 × N_A
= 6 × 10^{23}

(52) Answer : (3)
Hint:
One mole of $Ba_3(PO_4)_2$ contains 3 mole of Ba, 2 mole of P and 8 mole of O.
Solution:
In 1 mole of $Ba_3(PO_4)_2$ there is 8 moles of O = 128 g of oxygen, so 0.1 mole of $Ba_3(PO_4)_2$ contains 12.8 g of oxygen.

(53) Answer : (3)
Solution:

$$\underset{2\text{ moles}}{2C_6H_6} + \underset{7\text{ moles}}{7O_2} + \rightarrow 4\ CO_2 + 6\ H_2O$$

For 2 moles of C_2H_6 = 7 moles of O_2 required.

$$\boxed{\text{i.e., for } 60\text{ g of } C_2H_6 = 7\times22.4\text{ L of } O_2 \text{ at STP required}}$$

(54) Answer : (2)
Hint:

Molality $= \frac{\text{Moles of solute}}{\text{Mass of solvent (kg)}}$

Solution:
In one molal aqueous solution, 1 kg water is solvent and one mole of solute is present
Mole fraction of solvent (water) = $\frac{1000/18}{1000/18+1}$
= $\frac{55.56}{56.56} = 0.98$

(55) Answer : (3)
Hint:
$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$
Solution:
Molarity of $Al_2(SO_4)_3 = \frac{34.2/342}{500/1000}$ = 0.2 M
Molarity of Al^{3+} ions = 0.2 × 2 = 0.4 M

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and molarity of SO_4^{2-} ions = 0.2 × 3 = 0.6 M

(56) **Answer :** (1)
Hint:
$N_2 + 3H_2 \rightarrow 2NH_3$
Solution:
Moles of N_2 = $\frac{14}{28} = 0.5$
Moles of H_2 = $\frac{6}{2} = 3$
According to stoichiometry 0.5 moles of N_2 will react with 1.5 moles of H_2. So, H_2 is in excess and N_2 is limiting reagent.
Mole of ammonia produced = 0.5 × 2 = 1
Mass of ammonia produced = 17 g

(57) **Answer :** (1)
Solution:
Molarity = $\frac{3.011\times10^{21}\times1000}{6.022\times10^{23}\times200}$
$= \frac{10^{-2}\times5}{2}$
= $2.5\times10^{-2}M$

(58) **Answer :** (4)
Solution:
Zeros between two non zero digits are significant.

(59) **Answer :** (3)
Solution:
1 mole of atomic H = 6.022 × 10^{23} hydrogen atoms.

(60) **Answer :** (4)
Hint:
Average atomic mass $= \frac{\sum \%\text{ of isotope}\times\text{atomic mass}}{100}$
Solution:
Average atomic mass $= \frac{60\times60+40\times62}{100} = 60.8\text{ amu}$

(61) **Answer :** (2)
Solution:

A	+	2B	→	3C
2		2		0
1		0		3

3 moles of c is formed.
1 mole of A is left after the reaction.

(62) **Answer :** (4)
Solution:
The prefix micro is used for 10^{-6} multiple.

(63) **Answer :** (2)
Solution:
Molarity = $\frac{m\%\times d\times10}{M_{H_2SO_4}}$
$M = \frac{49\times1.09\times10}{98} = 5.45\text{ M}$

(64) **Answer :** (2)
Solution:
Empirical formula is the simplest whole number ratio of elements present in a compound.

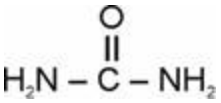
Element	%	Mole	Ratio
C	40	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	6.67	$\frac{6.67}{1} = 6.67$	$\frac{6.67}{3.33} = 2$
O	53.3	$\frac{53.3}{16} = 3.33$	$\frac{3.33}{3.33} = 1$

Empirical formula is CH_2O

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(65) **Answer :** (3)
Hint:
After dilution, number of moles of solute remains same, hence
$M_1V_1 = M_2V_2$
Solution:
$M_1V_1 = M_2V_2$
$V_2 = \frac{M_1V_1}{M_2} = \frac{100\times5}{0.25} = 2000$ mL
Water added = 2000 – 100 = 1.9 L

(66) **Answer :** (1)
Solution:
Structure of urea :



Mole of urea = $\frac{5.4}{60} = 0.09$
Number of hydrogen atoms = $0.09 \times 4 \times 6.022 \times 10^{23}$
$= 2.1679 \times 10^{23}$
$\simeq 2.168 \times 10^{23}$

(67) **Answer :** (4)
Hint:
Number of mole(s) $= \frac{\text{Number of molecules}}{N_A}$
Solution:
Mole of sugar $= \frac{3.01\times10^{23}}{6.02\times10^{23}} = 0.5$
Mole fraction of sugar $= \frac{0.5}{0.5+24.5} = 0.02$

(68) **Answer :** (1)
Solution:
Precision refers to closeness of various measurements for the same quantity

(69) **Answer :** (2)
Solution:
% of Cu and O is same from different source and follows law of constant proportion.

(70) **Answer :** (2)
Solution:
$2H_2(g) + O_2(g) \rightarrow 2H_2O$
2 Volume 1 Volume

(71) **Answer :** (1)
Solution:
(1) Number of molecules
$\left(H_2\right) = \frac{15}{22.4} \times N_A = 0.669N_A$
(2) Number of molecules
$\left(N_2\right) = \frac{5}{22.4} \times N_A = 0.223N_A$
(3) Number of molecules
$\left(H_2\right) = \frac{0.5}{2} \times N_A = 0.25N_A$
(4) Number of molecules
$\left(O_2\right) = \frac{10}{32} \times N_A = 0.3125N_A$

(72) **Answer :** (4)
Hint:
Mole of molecule/atom = $\frac{\text{Mass of molecule / atom}}{\text{Molar mass of molecule / atom}}$
Solution:

Mole of He = $\frac{2x}{4}$

Mole of N_2 = $\frac{x}{28}$

$\frac{\text{Mole of He}}{\text{Mole of } N_2} = \frac{\frac{2x}{4}}{\frac{x}{28}} = \frac{2x}{4} \times \frac{28}{x} = 14:1$

(73) Answer : (2)
Hint:

Number of moles = $\frac{Given\ mass\ (in\ g)}{Molar\ mass} = \frac{N}{N_A}$

Solution:

Number of molecules left = $\left(\frac{220 \times 10^{-3}}{44} \times 6.022 \times 10^{23}\right) - 10^{21}$

= $3.011 \times 10^{21} - 10^{21}$

= 2.011×10^{21}

(74) Answer : (1)
Solution:

Mass percent = $\frac{5}{5+15} \times 100 = \frac{5}{20} \times 100 = 25\%$

(75) Answer : (3)
Solution:

$4\ \text{mol of } NH_3 \equiv 5\ \text{mol of } O_2$

$\therefore 2\ \text{mol of } NH_3 \equiv \frac{5}{4} \times 2 = 2.5\ \text{mol of } O_2\ (\text{required})$

Available moles of $O_2 = 2$, Hence, O_2 is limiting reagent.

$\Rightarrow 2\ \text{mol of } O_2 \text{ reacts with } = \frac{4}{5} \times 2 = 1.6\ \text{mol of } NH_3$

Excess amount of $NH_3 = 2 - 1.6 = 0.4\ \text{mol}$

Volume of excess NH_3 at STP = $0.4 \times 22.4 = 8.96\ \text{L}$

Moles of NO produced = $1.6\ \text{mol}$

Moles of H_2O produced = $2.4\ \text{mol}$

Hence, molecules of H_2O = 2.4 N_A

(76) Answer : (2)
Solution:

$C_3H_8 + 5O_2 \rightarrow 3\,CO_2 + 4H_2O$

moles of $C_3H_8 = \frac{44.8}{22.4} = 2\ \text{mol}$

For 1mole of $C_3H_8 \Rightarrow 5$ moles of O_2 required

For 2moles of $C_3H_8 \Rightarrow 10$moles of O_2 required

weight of $O_2 = 10 \times 32 = 320\ \text{g}$

(77) Answer : (4)
Solution:

Number of moles of atoms = $\frac{3.01 \times 10^{24}}{6.02 \times 10^{23}} = 5$

$\therefore$ Moles of molecules = $\frac{5}{3}$ (for triatomic molecule)

Molar mass = $\frac{3}{5} \times 80 = 48$ g/mol

(78) Answer : (3)
Solution:

$\frac{^\circ C}{5} = \frac{^\circ F - 32}{9}$

$\frac{65}{5} = \frac{^\circ F - 32}{9}$

$\therefore {}^\circ F = 149$

(79) Answer : (2)
Solution:

Mass percentage of oxygen

$= \frac{\text{Weight of oxygen}}{\text{Weight of } H_2SO_4} \times 100$

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= $\frac{64}{98}$ × 100 = 65.31%

(80) **Answer :** (2)
Hint:
1 P^{3-} ion has 18 electrons.
Solution:
1 mol P^{3-} ions have 18 N_A electrons

(81) **Answer :** (4)
Solution:
$\mathrm{M} = \frac{\%\text{ by mass} \times \text{density} \times 10}{\text{molar mass}}$ = $\frac{98 \times 1.8 \times 10}{98} = 18\,M$
Now, $\mathrm{M_1V_1 = M_2V_2}$
$\Rightarrow \;\; 18\,V_1 = 0.2 \times 500 \;\Rightarrow\; V_1 = 5.55\,ml$

(82) **Answer :** (4)
Solution:
$\frac{1}{4}$ mole weighs 66.75 g
⇒ 1 mole weighs 4 × 66.75 g = 267 g
Hence molecular mass = 267 u

(83) **Answer :** (4)
Solution:
$X_{solute} = \frac{\frac{42.5}{85}}{\frac{42.5}{85} + \frac{57.5}{18}} = \frac{0.5}{0.5+3.19} = 0.135$

(84) **Answer :** (1)
Solution:
wt. of glucose = 40 g
wt. of solution = 1.2 × 100 = 120 g
ppm of glucose = $\frac{40}{120} \times 10^6 = 3\,.3 \times 10^5$ ppm

(85) **Answer :** (1)
Solution:
$\frac{\text{atoms of } H_2}{\text{atoms of } CO_2} = \frac{2 \times n_{H_2}}{3 \times n_{CO_2}}$
$= \frac{2}{3} \times \frac{w_{H_2}}{2} \times \frac{44}{w_{CO_2}} = \frac{44}{3} \times \frac{1}{11} = \frac{4}{3}$

(86) **Answer :** (3)
Hint:
Molality = $\frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$
Solution:
18 g glucose in 180 g water
$\frac{18/180}{180} \times 1000 = \frac{5}{9}$ m
34.2 g sucrose in 360 g water
$\frac{34.2/342}{360} \times 1000 = \frac{5}{18}$ m
5.85 g NaCl in 18 g water
$\frac{5.85/58.5}{18} \times 1000 = \frac{50}{9}$ m
4.6 g C_2H_5OH in 36 g water
$\frac{4.6/46}{36} \times 1000 \Rightarrow \frac{25}{9}$ m

(87) **Answer :** (3)
Solution:
An empirical formula represents the simplest whole number ratio of various atoms present in a compound. Whereas the molecular formula shows the exact number of different types of atoms present in a molecule of a compound.

(88) **Answer :** (1)
Hint:

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Molar mass = $\frac{\text{Given mass}}{\text{Number of moles}}$

Solution:

Suppose the molar masses of X and Y are x g mol^{-1} and y g mol^{-1} respectively
Molar mass of X_2Y = 2x + y = 100g
Molar mass of X_3Y_4 = 3x + 4y = 300g
On solving above equations for x and y
x = 20 u, y = 60 u

(89) **Answer :** (3)
Solution:
Atoms of different elements combine in a fixed ratio to form compound.

(90) **Answer :** (4)
Hint:
One molecule of Biomolecule should contain one atom of Zn.
Solution:
$\%\ \text{of Zn} = \frac{\text{Number of Zn atom} \times 65.38}{\text{wt. of biomolecule}} \times 100$
$0.75 = \frac{1 \times 65.38}{\text{MW}} \times 100$
$\therefore$ Minimum molar mass of biomolecule
$= \frac{65.38 \times 100}{0.75}$
= 8717.33 g/mol

BOTANY

(91) **Answer :** (2)
Solution:
Lysosome – Part of endomembrane system and rich in hydrolytic enzyme.
Peroxisome – Single membrane bound minute vesicles containing enzymes.
Mesosome – Membrane extension in prokaryotes

(92) **Answer :** (3)
Hint:
Glycocalyx differs in the composition and thickness among different bacteria.
Solution:
The outermost layer of cell envelope in bacteria, *i.e.*, glycocalyx could be a loose sheath called the slime layer in some, while in others, it may be thick and tough, called the capsule.

(93) **Answer :** (4)
Solution:
Small fragments which are formed beyond the secondary constriction are called satellite.

(94) **Answer :** (1)
Solution:
80S ribosome consists of 60S and 40S ribosomal subunits.

(95) **Answer :** (3)
Hint:
It is involved in uptake and release of Ca^{2+} ions in muscle cells.
Solution:
Sarcoplasmic reticulum is the SER of muscle cells.

(96) **Answer :** (2)
Hint:
These are rich in hydrolytic enzymes which get activated at acidic pH.
Solution:
Lysosomes are the single membrane bound, polymorphic cell organelles which are associated with intracellular digestion of food materials. Lysis of old or dead organelles and ephagy of undigested substances are also carried out by lysosomes. But conversion of fats into carbohydrates is a function of glyoxysomes.

(97) **Answer :** (2)
Solution:

In the given diagram of mitochondria, label A is outer membrane, label B is crista, label C is matrix and label D is inner membrane.

(98) Answer : (2)
Solution:
Centrosome duplicates during S-phase of cell cycle.
9 + 2 organisation is present in flagella and cilia of eukaryotic cell.
Anything less than a complete structure of a cell does not ensure independent living. Thus, cell is the fundamental, structural and functional unit of all living organism.

(99) Answer : (2)
Solution:
Small bristle like fibres on bacteria are called fimbriae.
Fimbriae are helpful in attachment to the surface like host tissue or substratum.

(100) Answer : (2)
Solution:
Presence of secondary constriction in chromosome gives the appearance of a small fragment called the satellite.

(101) Answer : (3)
Solution:
Centriole has an organisation like the cartwheel. It is surrounded by amorphous pericentriolar materials.

(102) Answer : (2)
Hint:
Chromatophores are membranous extensions into the cell of some prokaryotes.
Solution:
Chromatophores have photosynthetic pigments and the membrane has lipids and proteins similar to that of plasma membrane. These are found in photosynthetic prokaryotes, like cyanobacteria and purple bacteria.

(103) Answer : (2)
Solution:
Hint: The two ribosomal subunits associate with each other and is known as dimer.
Sol. : The two ribosomal subunits remain united with each other due to a specific concentration of the Mg^{2+} ions.

(104) Answer : (3)
Solution:
Lysosomes have hydrolytic enzymes which are active at acidic pH.

(105) Answer : (4)
Hint:
Amyloplast, elaioplast and aleuroplast are the types of leucoplast.
Solution:
Elaioplast stores fats and oils, aleuroplast stores proteins and amyloplast is starch-containing leucoplast.
Chromoplasts are yellow, orange or reddish in colour because of the presence of carotenoid pigments.

(106) Answer : (2)
Solution:
Oxidative phosphorylation occurs in mitochondria.

(107) Answer : (2)
Solution:
Theodore Schwann studied different types of animal cells and reported that cells had a thin outer layer which is today known as the ‘plasma membrane’.

(108) Answer : (4)
Hint:
Cell wall gives shape to the cell and protects the cell from mechanical damage and infection. It also helps in cell-to-cell interaction and provides barrier to undesirable macromolecules.
Solution:
In a young plant cell, the primary cell wall is capable of growth, which gradually diminishes as the cell matures.

(109) Answer : (3)
Hint:
Chloroplast is capable of self duplication. It can synthesize some of its proteins but receives other proteins from cytoplasm.
Solution:
Chloroplast and mitochondria both are semi-autonomous cell organelles not completely autonomous. They are dependent on nucleus for their some proteins.

(110) Answer : (4)

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Solution:
Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm.

(111) Answer : (3)
Solution:
The cell theory given by Schleiden and Schwann did not explain how new cells were formed.
Rudolf Virchow first explained that cells divided and new cells are formed from pre-existing cells, *i.e*., *Omnis cellula-e cellula*.

(112) Answer : (3)
Solution:
Prokaryotes have 70S ribosomes that are made up of 50S and 30S subunits.

(113) Answer : (3)
Solution:
Microtubules help in cell plate formation during plants cytokinesis.
Formation of cleavage furrow during animal cell division occur with the help of microfilaments.

(114) Answer : (2)
Solution:
Yeast is a fungus and fungi are eukaryotes.

(115) Answer : (4)
Solution:
Vacuoles are membrane bound space in cytoplasm but gas vacuole is membraneless, found in prokaryotes. It is not a part of endomembrane system.
Endomembrane system includes ER, Golgi complex, lysosome and vacuoles.

(116) Answer : (3)
Solution:
A disc-shaped structure is found over centromere through which spindle fibres are attached called kinetochore.

(117) Answer : (4)
Hint:
Microtubules form spindle fibres in eukaryotic cells.
Solution:
Microtubules are made up of tubulin protein.

(118) Answer : (4)
Hint:
Camillio Golgi first observed densely stained reticular structures near the nucleus. These were later named Golgi bodies after him.
Solution:
Golgi bodies consist of cisternae that are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face. They are in close association with ER.

(119) Answer : (3)
Hint:
Nucleolus is the site of active r-RNA synthesis.
Solution:
Prokaryotes lack nucleolus & nucleus thus it will be absent in purple & green sulphur bacteria.

(120) Answer : (4)
Hint:
Metacentric chromosome is isobrachial chromosome. Sub-metacentric chromosome, acrocentric chromosome and telocentric chromosome have submedian, subterminal and terminal centromere.
Solution:
In sub-metacentric chromosome, centromere is present slightly away from the middle.
Acrocentric chromosome have one extremely short arm and one very long arm. Telocentric chromosomes have terminal centromere. In metacentric chromosomes, both arms are equal.

(121) Answer : (2)
Hint:
Endocytosis is intake of materials in the form of carrier vesicles formed by invagination of small regions of plasma membrane.
Solution:
Functions like cell growth, formation of intercellular junctions, secretion etc. are due to fluid nature of plasma membrane.

(122) Answer : (3)
Hint:
The plasmid DNA confers certain unique phenotypic characters, such as resistance to antibiotics, to some bacteria.

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Solution:
Bacteria are prokaryotes and prokaryotes have double stranded circular DNA as genetic material.

(123) Answer : (2)
Hint:
The network of tiny tubular structures scattered in the cytoplasm represents ER.
Solution:
The biomolecules synthesized in ER reach the *cis* face of Golgi complex. Within the Golgi complex the biomolecules is transported to the *trans* face and ultimately release from this concave face.

(124) Answer : (2)
Hint:
Grana is the stack of thylakoids.
Solution:
In chloroplasts, there are flat membranous tubules called the stroma lamellae connecting the thylakoids of the different grana.

(125) Answer : (3)
Solution:
Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the organelles of endomembrane system, these are not considered as part of the endomembrane system.

(126) Answer : (3)
Solution:
Inner mitochondrial membrane is not continuous and has 20% lipids and 80% proteins. It Is rich in cardiolipin.

(127) Answer : (2)
Solution:
The main arena of cellular activities in plant and animal cells is cytoplasm.

(128) Answer : (3)
Solution:
Chromatin is composed of DNA, basic protein histone, non-histone proteins and RNA.

(129) Answer : (1)
Solution:
A phospholipid is an amphipathic molecule, thus, it can exist as a stable boundary between two aqueous compartments.

(130) Answer : (4)
Solution:
In higher plants centrioles are absent.

(131) Answer : (3)
Hint:
Microbodies are single membrane bound minute vesicles.
Solution:
Microbodies contain various enzymes, are present in both plant and animal cells. These are in the form of minute vesicles.

(132) Answer : (2)
Hint:
Prokaryotic flagella differs structurally from eukaryotic flagella.
Solution:
In eukaryotes, flagella have 9 + 2 organisation of microtubules. Both eukaryotic & prokaryotic flagella are functionally same.

(133) Answer : (2)
Solution:
Nucleus was discovered by Robert Brown.

(134) Answer : (4)
Solution:
The shape of mesophyll cell is round and oval.

(135) Answer : (3)
Solution:
Middle lamella is composed of calcium pectate and magnesium pectate mainly and holds the two neighbouring cells together.

ZOOLOGY

(136) Answer : (2)

Hint:

Inorganic compounds are found in filtrate.

Solution:

Acid-insoluble pool contains only lipids, polysaccharides, proteins and nucleic acids. Filtrate is other name given to acid-soluble pool whereas retentate is other name for acid-insoluble pool. Sulphates and phosphates are found in acid-soluble fraction. Only organic compounds are considered as biomolecules.

(137) Answer : (1)

Solution:

Epithelial tissue has free surface, which faces either a body fluid or the outside environment and thus provides a covering or a lining for some parts of the body. Epithelial tissue is present in the inner surface of hollow organs like bronchioles and fallopian tubes.

(138) Answer : (4)

Solution:

In aerolar tissue, fibroblasts, mast cells and macrophages are present.

Goblet cells secrete mucus and these are present in epithelial tissue.

(139) Answer : (1)

Solution:

'**X**' is a ligament and it is a type of dense regular connective tissue in which the collagen fibres are present in rows between many parallel bundle of fibres. '**Y**' is adipose tissue and it is a type of loose connective tissue that is located mainly beneath the skin. Cells of this tissue are specialised to store fats. The excess of nutrients which are not used immediately are converted into fats and stored in this tissue.

(140) Answer : (2)

Solution:

The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has microvilli. The simple cuboidal epithelium is composed of a single layer of cube-like cells. Their nuclei are present at the centre.

(141) Answer : (3)

Solution:

Alveoli of lungs are lined by simple squamous epithelium.

(142) Answer : (2)

Solution:

I_2 molecules get trapped in helices of starch to give blue black color. Both starch and glycogen are homopolymers of glucose.

(143) Answer : (2)

Solution:

The tissue is organised in specific proportion and pattern to form an organ like stomach, lung, heart and kidney

(144) Answer : (4)

Solution:

Skin is lined by stratified epithelium.

Diffusion, absorption and filtration are functions of simple epithelium.

(145) Answer : (2)

Solution:

Element	% weight of human body
Hydrogen	9.5
Carbon	18.5
Oxygen	65.0
Calcium	1.5

(146) Answer : (2)

Solution:

Hint: It is a homopolysaccharide

Sol.: The exoskeleton of an arthropod is made up of chitin. Chitin is a homopolysaccharide that is made up of repeating units of N-acetyl glucosamine. These repeating units are held together by β(1 → 4) glycosidic linkage. It is also the primary component of cell wall in fungi.

(147) Answer : (1)

Solution:

Fibres and fibroblasts are compactly packed in the dense connective tissues. Orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular connective tissues respectively.

(148) Answer : (4)

Solution:

Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

(149) Answer : (4)

Solution:

Stratified epithelium has more than one layer. Due to the presence of mucus secreting goblet cells, columnar epithelium is also known as simple columnar glandular epithelium.

(150) Answer : (4)

Solution:

Tissues are organised in specific proportions and patterns to form an organ.
All types of tissues are present in organs such as stomach, heart, lung and kidney.

(151) Answer : (3)

Solution:

Primary metabolites take part in normal physiological processes.

(152) Answer : (4)

Solution:

Polymers can be broken into monomers by addition of water (hydrolysis). Starch, glycogen and cellulose are homopolymers of glucose.

(153) Answer : (2)

Hint:

Feature of cardiac muscle fibres

Solution:

Smooth muscle fibres taper at both ends (fusiform) and do not show striations. Cell junctions hold them together. Smooth muscles are involuntary in nature. Cardiac muscle fibres are branched. Skeletal muscle fibres and cardiac muscle fibres show striations.

(154) Answer : (2)

Hint:

The cells that secrete collagen or elastin are absent in blood.

Solution:

In all connective tissues, except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissues.

(155) Answer : (1)

Solution:

Element	**% Weight of**	
	Earth's crust	**Human body**
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

(156) Answer : (1)

Hint:

Water is the most abundant chemical in living tissue.

Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(157) Answer : (3)

Solution:

Mast cells produce histamine.

(158) Answer : (4)

Solution:

The acid-insoluble fraction contains compounds with molecular weight higher than ten thousand daltons, except lipids.

(159) Answer : (1)

Solution:

Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products.
Insulin and thyroxine are secreted from endocrine glands.

(160) Answer : (1)

Solution:

Cellulose is a polymeric substance and codeine is an alkaloid.

(161) Answer : (1)

Solution:

Goblet cells of the alimentary canal secrete mucus.

(162) Answer : (4)

Hint:

Compound epithelium

Solution:

Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

(163) Answer : (4)

Solution:

Cellulose is the most abundant structural polysaccharide in nature and is also a secondary metabolite categorised as a polymeric substance.

(164) Answer : (2)

Solution:

Inulin is a polymer of fructose.

(165) Answer : (4)

Solution:

Connective tissue is the most abundant and widely distributed tissue in the body of complex animals. It includes bone, cartilage, adipose tissue, areolar tissue and blood.

(166) Answer : (4)

Solution:

Average composition of cellular components is : Water > Proteins > Nucleic acids > Carbohydrates.

(167) Answer : (2)

Solution:

Smooth muscle fibres are spindle-shaped, uninucleated, and involuntary in nature.

(168) Answer : (2)

Solution:

In neural tissues, neurons coordinate between different organs through nerve impulses and it reacts or adjusts the body with respect to change in the environment.

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(169) Answer : (3)
Solution:
Tight junction helps to stop substances from leaking across a tissue.

(170) Answer : (3)
Solution:
Living organisms have also got inorganic elements and compounds in them.

(171) Answer : (3)
Solution:
Fibroblast, chondroblast and osteoblast all are immature cells.

(172) Answer : (4)
Solution:
Chondrocytes lie singly or in groups of two or four.

(173) Answer : (4)
Solution:
Squamous epithelium forms the innermost layer of walls of blood vessels (endothelium), air sac (alveoli) of the lungs.

(174) Answer : (1)
Hint:
They form the main skeletal framework of body.
Solution:
Bones interact with skeletal muscles having cylindrical fibres to bring about movements.
Bone marrow in some bones is the site of production of blood cells.

(175) Answer : (4)
Solution:
Right end in a polysaccharide chain of glycogen has free aldehyde group and is known as reducing end. Cellulose does not contain complex helices, as it is unbranched, hence cannot hold I_2.

(176) Answer : (2)
Solution:
Neural tissue – ectodermal origin
Muscular tissue – mesodermal origin
Both muscular tissue and neural tissue exhibit excitability

(177) Answer : (2)
Solution:
Chitin is a structural polysaccharide found in exoskeleton of insects.

(178) Answer : (4)
Solution:
Neuroglial cells are specialised cells found in brain and spinal cord supporting neurons. About more than 50% of all brain cells are neuroglial cells in volume in our body.

(179) Answer : (4)
Solution:
When dried tissue is fully burnt, the carbon compounds are oxidised to gaseous forms like CO_2 and water vapour and the remnant is called ‘ash’. This ash contains many inorganic elements like calcium, magnesium *etc*.

(180) Answer : (2)
Solution:
Visceral muscles of a human being such as those in the digestive tract are non-striated and smooth in appearance.

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